

ANTIBIOTICS
— ONLY IF:
FEVER + CXR INFILTRATE

SILENT CHEST
= OMINOUS

< 60% — ADMIT

PCO₂ = 40 mmHg

DANGER
IN ASTHMA

PERMIT
HIGH CO₂ —
LOW RATE

PREVENTS
INTUBATION

LOW
LOW RATE /
LOW VOLUME

I ARRIVE LATE

PERMIT
HIGH CO₂ —
LOW RATE /
LOW VOLUME

4-6 HRS

WHEN HOSE FAILS

Mg²⁺ SULFATE IV

ALBUTEROL

IPRATROPIUM

AMINOPHYLLINE

NARROW WINDOW / INFERIOR

1. Antibiotics locked

WHAT YOU SEE

A padlocked cabinet labeled 'Antibiotics — only if fever + CXR infiltrate'.

WHAT IT ENCODES

Most asthma exacerbations are triggered by viral URIs, allergens, or non-adherence — not bacterial infection — so antibiotics are NOT routine. Reserve them for clear bacterial pneumonia (focal infiltrate + fever + purulent sputum) or proven bacterial sinusitis. Routine CXR is not needed for a typical exacerbation either; obtain it only if you suspect pneumonia, pneumothorax, or the patient fails to respond.

BOARD TRAP

Do not give empiric antibiotics for an asthma exacerbation without evidence of bacterial pneumonia — purulent-looking sputum in asthma is usually eosinophil-laden, not bacterial.

2. Silent chest / PEF <60% admit

WHAT YOU SEE

A skull-and-crossbones bell with 'Silent chest = ominous' and a 'PEF <60% → admit' gauge.

WHAT IT ENCODES

A silent chest means airflow is too low to generate wheeze — a sign of life-threatening obstruction and impending arrest. Severity is graded by PEF: ≥50-75% predicted = moderate, 33-50% = acute severe, <33% (or silent chest, cyanosis, bradycardia, exhaustion, SpO₂ <92%) = life-threatening. PEF/FEV₁ <60% predicted that fails to improve warrants admission.

BOARD TRAP

Absence of wheeze is NOT reassuring — a quiet chest in a distressed asthmatic signals too little air movement to wheeze, i.e., near-fatal obstruction.

3. Firefighters at flames (inflamed lungs)

WHAT YOU SEE

Firefighters battling flames shaped like inflamed bronchial lungs.

WHAT IT ENCODES

The acute attack is driven by bronchial smooth-muscle spasm, mucosal edema, and mucus plugging on a background of type-2 (eosinophilic) airway inflammation. The therapeutic strategy mirrors the firefight: bronchodilators (SABA ± ipratropium) relax muscle immediately while systemic steroids extinguish the underlying inflammation.

4. Normal/rising PCO₂ = danger

WHAT YOU SEE

A PCO₂ gauge at 40 mmHg with 'DANGER in asthma' beside a red alarm light.

WHAT IT ENCODES

Early in an attack the tachypneic patient blows off CO₂ → respiratory ALKALOSIS with low PaCO₂ (the expected ABG). As fatigue sets in, the PaCO₂ climbs back toward 'normal' (40) and then high — this normalization/rise signals the patient can no longer sustain the work of breathing and is heading for respiratory failure. A rising PaCO₂ with falling pH is an indication to consider intubation.

BOARD TRAP

A 'normal' PaCO₂ in a tachypneic, distressed asthmatic is a red flag, not reassurance — it means fatigue and impending failure, not improvement.

5. Mg²⁺ sulfate cannon

WHAT YOU SEE

A cannon labeled 'Mg²⁺ sulfate IV — when hose fails'.

WHAT IT ENCODES

IV magnesium sulfate (typically 2 g over 20 min) is a smooth-muscle relaxant used as adjunctive rescue for severe/life-threatening exacerbations not responding to first-line nebulized SABA + ipratropium + steroids. It modestly reduces hospital admissions in severe attacks.

BOARD TRAP

Magnesium is adjunct/rescue, not first-line — SABA, ipratropium, and systemic steroids come first; reaching for Mg before them is the wrong sequence.

6. Albuterol firehose (SABA)

WHAT YOU SEE

The firefighter sprays a hose labeled 'Albuterol'.

WHAT IT ENCODES

Inhaled SABA (albuterol/salbutamol) is the first-line acute bronchodilator for every exacerbation — give early, repeated, and via nebulizer or MDI-spacer with continuous dosing in severe cases. It acts within minutes on β₂ receptors to relax airway smooth muscle.

BOARD TRAP

Side effects of high-dose β₂-agonist: tachycardia, tremor, hypokalemia, and lactic acidosis — don't misread the lactate or low K as a separate disease process.

7. Ipratropium hose (SAMA)

WHAT YOU SEE

A second hose labeled 'Ipratropium' joining the spray.

WHAT IT ENCODES

Add nebulized ipratropium (a SAMA/anticholinergic) to SABA in moderate-to-severe exacerbations — the combination produces greater bronchodilation and fewer admissions than SABA alone. It is an ED adjunct, not a maintenance controller.

BOARD TRAP

Ipratropium is for the acute attack only — it has no role in chronic asthma maintenance (that's tiotropium/LAMA at higher steps).

8. Intubation prevention

WHAT YOU SEE

A panel 'Prevents intubation' with 'permit high CO₂ / low rate / low volume'.

WHAT IT ENCODES

Maximal pharmacotherapy (continuous SABA, ipratropium, steroids, IV Mg $\pm$ heliox) aims to avoid intubation, which is high-risk in asthma. If ventilation is required, use a lung-protective, low-rate, low-tidal-volume strategy with prolonged expiratory time to allow full exhalation and permissive hypercapnia.

BOARD TRAP

Do not hyperventilate the intubated asthmatic — high respiratory rates prevent complete exhalation, causing breath-stacking, dynamic hyperinflation (auto-PEEP), hypotension, and barotrauma/pneumothorax.

9. Monitor / ventilator settings

WHAT YOU SEE

A bedside monitor with 'permit high CO₂ — low rate, low volume'.

WHAT IT ENCODES

Ventilator settings in severe asthma maximize expiratory time (low rate ~8-12, low tidal volume ~6 mL/kg, high inspiratory flow, minimal/zero applied PEEP) to combat air-trapping. Tolerate hypercapnia (pH down to ~7.2) rather than raising the rate. Sudden hypotension on the vent = check for auto-PEEP/tension pneumothorax (disconnect to exhale).

BOARD TRAP

If a ventilated asthmatic suddenly drops their blood pressure, the first move is to DISCONNECT the circuit to let trapped air escape (auto-PEEP) and rule out tension pneumothorax — not to give more fluids blindly.

10. Steroids 'I arrive late' (4-6 hrs)

WHAT YOU SEE

A sleepy ghost figure 'I arrive late' in a chair with an hourglass '4-6 hrs'.

WHAT IT ENCODES

Systemic corticosteroids (oral prednisone ~40-50 mg or IV methylprednisolone) reduce airway inflammation and relapse but take 4-6 hours to act — so give them EARLY in every moderate-to-severe attack rather than waiting. Oral and IV are equivalent in efficacy; a typical course is 5-7 days with no taper needed for short courses.

BOARD TRAP

Steroids are not bronchodilators and won't relieve symptoms immediately — but the delay is exactly why you start them at the first hour, not after bronchodilators fail.

11. Epinephrine in trash

WHAT YOU SEE

A bottle of epinephrine tossed in a trash can with a red X.

WHAT IT ENCODES

IM epinephrine is the treatment for ANAPHYLAXIS, not routine asthma — for an ordinary exacerbation, inhaled SABA delivers selective β_2 bronchodilation without epinephrine's systemic α/β_1 effects. Epinephrine is reserved for asthma only when anaphylaxis is the trigger or as a last resort in peri-arrest.

BOARD TRAP

Do not reach for epinephrine in a routine asthma attack — but DO give it immediately if the attack is part of anaphylaxis (urticaria, angioedema, hypotension after an allergen).

12. Theophylline crossed out

WHAT YOU SEE

A bottle labeled theophylline-type drug with a red X near the cannon.

WHAT IT ENCODES

Theophylline/aminophylline is largely abandoned in acute asthma — it adds no benefit over standard therapy and has a narrow therapeutic window with dangerous toxicity (arrhythmias, seizures, vomiting), worsened by interactions (macrolides, ciprofloxacin, cimetidine inhibit its metabolism).

BOARD TRAP

Theophylline is NOT recommended in acute exacerbations — and watch for toxicity precipitated by adding a macrolide or fluoroquinolone, which raises theophylline levels.